

Assignment 3: Processing for DoorDash Delivery Data

S3 Buckets:

- 1. Created Json file, '2024-03-10-raw_data_json.json' with json data

```
{
  "2024-03-10-raw_data_json.json": [
    {
      "id": 1,
      "status": "delivered",
      "amount": 20.5,
      "date": "2024-03-09"
    },
    {
      "id": 2,
      "status": "cancelled",
      "amount": 15.0,
      "date": "2024-03-09"
    },
    {
      "id": 3,
      "status": "order placed",
      "amount": 22.5,
      "date": "2024-03-09"
    },
    {
      "id": 4,
      "status": "delivered",
      "amount": 19.5,
      "date": "2024-03-09"
    },
    {
      "id": 5,
      "status": "cancelled",
      "amount": 18.0,
      "date": "2024-03-09"
    },
    {
      "id": 6,
      "status": "delivered",
      "amount": 23.5,
      "date": "2024-03-09"
    },
    {
      "id": 7,
      "status": "order placed",
      "amount": 20.0,
      "date": "2024-03-09"
    },
    {
      "id": 8,
      "status": "delivered",
      "amount": 25.5,
      "date": "2024-03-09"
    },
    {
      "id": 9,
      "status": "delivered",
      "amount": 21.5,
      "date": "2024-03-09"
    },
    {
      "id": 10,
      "status": "cancelled",
      "amount": 17.5,
      "date": "2024-03-09"
    }
  ]
}
```

- 2.
- 3. Created Buckets 'doordash-landing-zn-gds-assign-3' and 'doordash-target-zn-gds-assign-3'

☐	doordash-landing-zn-gds-assign-3	US East (N. Virginia) us-east-1	Bucket and objects not public	March 10, 2024, 18:44:21 (UTC+05:30)
☐	doordash-target-zn-gds-assign-3	US East (N. Virginia) us-east-1	Bucket and objects not public	March 10, 2024, 18:45:26 (UTC+05:30)

- 4.

SNS Topic:

- 1. Create a SNS topic

Amazon SNS > Topics > s3-noti

s3-noti

EditDeletePublish message

Details

Name

s3-noti

ARN

arn:aws:sns:us-east-1:533267408419:s3-noti

Type

Standard

Display name

-

Topic owner

533267408419

<SubscriptionsAccess policyData protection policyDelivery policy (HTTP/S)Delivery status loggingEncryptionTags>

- 2.
- 3. Create a subscription email and verify it.

Amazon SNS > Topics > s3-noti > Subscription: 5eada5e9-7d3a-473a-a0e7-43a829d6867d

Subscription: 5eada5e9-7d3a-473a-a0e7-43a829d6867d

Edit Delete

Details

ARN arn:aws:sns:us-east-1:533267408419:s3-noti:5eada5e9-7d3a-473a-a0e7-43a829d6867d	Status Confirmed
Endpoint gourish.ganta777@gmail.com	Protocol EMAIL
Topic s3-noti	
Subscription Principal arn:aws:iam::533267408419:root	

4.

Lambda:

1. Create a lambda and add trigger to the s3 bucket for create object.

Lambda > Functions > aws-lambda-gds-assign3

aws-lambda-gds-assign3

Throttle Copy ARN Actions

Function overview

Export to Application Composer Download

Diagram Template

The diagram shows a box for the function 'aws-lambda-gds-assign3' with a 'Layers' section indicating 3 layers. Below it, an 'S3' bucket icon is shown with a line connecting it to the function box, indicating a trigger. There is also an '+ Add trigger' button.

+ Add destination

Description
-

Last modified
3 hours ago

Function ARN
arn:aws:lambda:us-east-1:533267408419:function:aws-lambda-gds-assign3

Function URL [Info](#)
-

2.

Trigger configuration [Info](#)

 **S3**
aws asynchronous storage

Bucket
Choose or enter the ARN of an S3 bucket that serves as the event source. The bucket must be in the same region as the function.

Bucket region: us-east-1

Event types
Select the events that you want to have trigger the Lambda function. You can optionally set up a prefix or suffix for an event. However, for each bucket, individual events cannot have multiple configurations with overlapping prefixes or suffixes that could match the same object key.

Prefix - optional
Enter a single optional prefix to limit the notifications to objects with keys that start with matching characters.

Suffix - optional
Enter a single optional suffix to limit the notifications to objects with keys that end with matching characters.

- 3.
4. Now add s3 and sns access roles for using the respective resources in the lambda function.

[IAM](#) > [Roles](#) > aws-lambda-gds-assign3-role-oyi76za2

aws-lambda-gds-assign3-role-oyi76za2 [Info](#)

Summary

Creation date March 12, 2024, 10:37 (UTC+05:30)	ARN  arn:aws:iam::533267408419:role/service-role/aws-lambda-gds-assign3-role-oyi76za2
Last activity  3 hours ago	Maximum session duration 1 hour

[Permissions](#) | [Trust relationships](#) | [Tags](#) | [Access Advisor](#) | [Revoke sessions](#)

Permissions policies (3) [Info](#)

You can attach up to 10 managed policies.

Filter by Type

☐	Policy name 	Type	Attached entities
☐	 AmazonS3FullAccess	AWS managed	4
☐	 AmazonSNSFullAccess	AWS managed	3
☐	 AWSLambdaBasicExecutionRole-742c2...	Customer managed	1

- 5.
6. Create layers for datetime and python-dotenv by adding the libraries in the python folder and zipping the files. Pandas library is already available in aws layers and can be used in the lambdas function.

Layers Info						Edit	Add a layer
Merge order	Name	Layer version	Compatible runtimes	Compatible architectures	Version ARN		
1	AWSSDKPandas-Python311	8	python3.11	x86_64	arn:aws:lambda:us-east-1:336392948345:layer:AWSSDKPandas-Python3		
2	date_and_time	1	python3.11	x86_64, arm64	arn:aws:lambda:us-east-1:533267408419:layer:date_and_time:1		
3	python-dotenv-library	1	python3.11	x86_64, arm64	arn:aws:lambda:us-east-1:533267408419:layer:python-dotenv-library:1		

7.

8. Lambda code:

- Defining necessary libraries.
- Read the bucket name and key name from the event. Read the json data and store the 'delivered' items in a dataframe.
- Store the dataframe in .csv file
- Connect with the target s3 bucket : doordash-target-zn-gds-assign-3. In the processed_data folder push the csv file created.
- Once its done, send the notification to the subscriptions provided in the topic.

Lambda_function.py

```
import json
import pandas as pd
import boto3
import io
from datetime import date
import os
from dotenv import load_dotenv

load_dotenv()

def lambda_handler(event, context):
    # Code added from CI CD
    input_bucket = event['Records'][0]['s3']['bucket']['name']
    input_key = event['Records'][0]['s3']['object']['key']
    s3 = boto3.client('s3')
    obj = s3.get_object(Bucket = input_bucket, Key = input_key)
    body = obj['Body'].read()
    json_dicts = body.decode('utf-8').split('\r\n')
    df = pd.DataFrame(columns = ['id', 'status', 'amount', 'date'])
    for line in json_dicts:
        py_dict = json.loads(line)
        if py_dict['status'] == 'delivered':
            df.loc[py_dict['id']] = py_dict
    df.to_csv('/tmp/test.csv', sep = ',')
    print('test.csv file created')
    try:
```

```

        date_var = str(date.today())
        file_name =
'processed_data/{}_processed_data.csv'.format(date_var)
    except:
        file_name = 'processed_data/processed_data.csv'
    lambda_path = '/tmp/test.csv'
    bucket_name = os.getenv('output_bucket')
    s3 = boto3.resource('s3')
    bucket = s3.Bucket(bucket_name)

    bucket.upload_file('/tmp/test.csv', file_name)

    # sns to deliver file processed request
    sns = boto3.client('sns')
    response = sns.publish(
        TopicArn=os.getenv('TopicArn'),
        Message="File {} has been formatted and filtered. Its been stored in
{} as {}".format(input_key,bucket_name,file_name)
    )

```

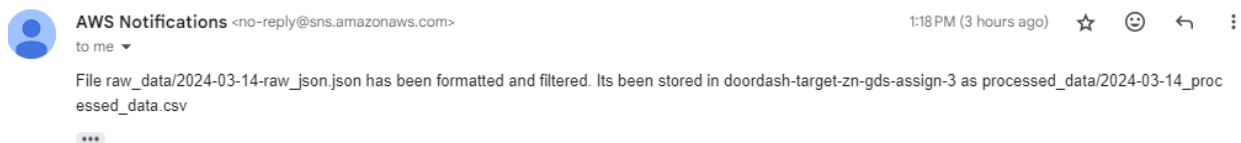
.env

```

output_bucket= doordash-target-zn-gds-assign-3
TopicArn= arn:aws:sns:us-east-1:533267408419:s3-noti

```

SNS result:



CI / CD Creation:

1. Create Github repository and push the code in the repository.
2. https://github.com/GantaGourish/doordash_project
3. Create different branches , test and main.

4. Use codeBuild to create new project and add options as shown below.

[Developer Tools](#) > [CodeBuild](#) > [Build projects](#) > [Create build project](#)

Create build project

Project configuration

Project name

A project name must be 2 to 255 characters. It can include the letters A-Z and a-z, the numbers 0-9, and the special characters - and _.

► **Additional configuration**
Description, Build badge, Concurrent build limit, tags

Source

Add source

Source 1 - Primary

Source provider

GitHub ▼

Repository

☒ Repository in my GitHub account

☐ Public repository

GitHub repository

X

- 5.

6.

GitHub repository

Q

https://github.com/GantaGourish/doorash_project.git

X

↺

https://github.com/<user-name>/<repository-name>

Connection status

You are connected to GitHub using OAuth.

Disconnect from GitHub

Source version - optional [Info](#)

Enter a pull request, branch, commit ID, tag, or reference and a commit ID.

main

► Additional configuration

Git clone depth, Git submodules

Primary source webhook events [Info](#)

Add filter group

Webhook - optional [Info](#) [↗](#)

☒ Rebuild every time a code change is pushed to this repository

Build type

☒ Single build

Triggers single build

☐ Batch build

Triggers multiple builds as single execution

Webhook event filter groups

A build is triggered if any filter group evaluates to true, which occurs when all the filters in the group evaluate to true.

Webhook event filter groups

A build is triggered if any filter group evaluates to true, which occurs when all the filters in the group evaluate to true.

Filter group 1

Remove filter group

Event type

Add one or more webhook event filter groups to specify which events trigger a new build. If you do not add a webhook event filter group, then a new build is triggered every time a code change is pushed to your repository.

PULL_REQUEST_MERGED X

► Start a build under these conditions - *optional*

► Don't start a build under these conditions - *optional*

Environment

Provisioning model Info [🔗](#)



On-demand

Automatically provision build infrastructure in response to new builds.



Reserved capacity

Use a dedicated fleet of instances for builds. A fleet's compute and environment type will be used for the project.

7. [Environment image](#)
8. For environment settings, use aws official documentation as required. Select correct image and runtime depending on the language and version you are using.
9. <https://docs.aws.amazon.com/codebuild/latest/userguide/available-runtimes.html>

Provisioning model Info [?](#)



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Use a dedicated fleet of instances for builds. A fleet's compute and environment type will be used for the project.

Environment image



Managed image

Use an image managed by AWS CodeBuild



Custom image

Specify a Docker image

Compute



EC2

Optimized for flexibility during action runs



Lambda

Optimized for speed and minimizes the start up time of workflow actions

Operating system

Amazon Linux



Runtime(s)

Standard



Image

aws/codebuild/amazonlinux2-x86_64-standard:5.0



Image version

Always use the latest image for this runtime version



☐ Use GPU-enhanced compute

10.

Buildspec

Build specifications



Insert build commands

Store build commands as build project configuration



Use a buildspec file

Store build commands in a YAML-formatted buildspec file

Buildspec name - *optional*

By default, CodeBuild looks for a file named buildspec.yml in the source code root directory. If your buildspec file uses a different name or location, enter its path from the source root here (for example, buildspec-two.yml or configuration/buildspec.yml).

buildspec.yml

11.

12. Use buildspec.yml file to provide instructions and commands to execute in vm image.

Buildspec.yml

```
version: 0.2
phases:
  install:
    runtime-versions:
      python: 3.11
    commands:
      - echo "Installing dependencies..."
      - pip install -r requirements.txt -t lib
  build:
    commands:
      - echo "Zipping deployment package..."
      - cd lib
      - zip -r9 ../deployment_package.zip .
      - cd ..
      - zip -g deployment_package.zip lambda_function.py
      - zip -g deployment_package.zip .env
  post_build:
    commands:
      - echo "Uploading to S3..."
      - aws s3 cp deployment_package.zip s3://zip-files-storage-lambda/
      - echo "Updating Lambda function via S3..."
      - aws lambda update-function-code --function-name
aws-lambda-gds-assign3 --s3-bucket zip-files-storage-lambda --s3-key
deployment_package.zip
      - echo "Deployment complete!"
```

Once you push the code in main branch, you can see logs in execution and verify the updated code in lambda function.

	Name ▾	Source provider	Repository	Latest build status	Description	Last Modified
○	doordash-project	GitHub	GantaGourish/doordash_project _	✔ Succeeded	-	4 hours ago

```
160
161 [Container] 2024/03/14 07:47:48.858544 Running command echo "Deployment complete!"
162 Deployment complete!
163
164 [Container] 2024/03/14 07:47:48.864812 Phase complete: POST_BUILD State: SUCCEEDED
165 [Container] 2024/03/14 07:47:48.864831 Phase context status code: Message:
166
```

